

Практикум C2

105 marks of the archive, nine techniques, thirteen tasks. Two topics that are really one: everything the archive asks about circular measure, and everything it asks about solids — which turns out to be cones, and cones come from sectors.

О практикуме

Две темы взяты целиком: `circular_measure` (32 блока, 99 баллов) и `solid_geometry` (8 блоков, 28). Порознь вторая не практикум — 28 баллов при нижней границе 60, — а вместе они одна тема и есть: конус в архиве получается из сектора, и других тел в теме нет.

Заданий

0

Раздел решений

включён

Интерактивная версия

kaggle.com/code/artemkonukhov/ib-math-praktikum-c2-circular-measure

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C2 — Radian measure, sectors, solids

105 marks of the archive, nine techniques, thirteen tasks. Two topics that are really one: everything the archive asks about circular measure, and everything it asks about solids — which turns out to be cones, and cones come from sectors.

The one idea

A radian is not another scale for angle. It is a **definition of angle as a length**:

$$\theta = \frac{\text{arc}}{\text{radius}}$$

Read that once more, because the whole topic is inside it. The angle *is* the number of radii that fit along the arc. Multiply both sides by r and you have $s = r\theta$ — not a formula to remember, but the same sentence written with a multiplication sign.

Everything else follows in one step. A full turn is $2\pi r/r = 2\pi$. A sector is the fraction $\theta/2\pi$ of the disc, so its area is $\frac{\theta}{2\pi} \cdot \pi r^2 = \frac{1}{2}r^2\theta$.

And this is exactly why **degrees may not be substituted**. What goes into these formulas is not an angle, it is a ratio of two lengths. A degree has no such ratio; it is $\frac{1}{360}$ of a convention.

The other idea

After that, the topic stops being about circles at all. Every remaining question is one move:

Cut the shaded shape into pieces whose measure you already know.

A segment is a sector minus a triangle. A logo is a rectangle minus two segments. A gutter is a segment plus a rectangle. The letter C is a sector minus a sector. That work happens on the diagram, before any button is pressed, and the

ladder below is a list of pieces, not of formulas.

How the checks work

They do not know the answers, and they do not know the formulas. Each check is handed the **boundary of the figure** — where the arcs run and where the straight edges run — and it measures:

```
verify_area('lc', qlc, *sector(4, 0, qlb))
```

is *"the shaded sector of the paper — the one with radius 4 and your angle — how big is it?"* The area comes out of a contour integral; $\frac{1}{2}r^2\theta$ appears nowhere, so any correct form of the answer passes: $\frac{3549\pi}{16}$, 696.844 and 697 are the same answer.

The checks run in both directions, and you will use both.

direction	what is known	what carries your answer
forward	the figure	the measurement
backward	the measurement	the figure

Backward looks like this:

```
verify_length('lb', 10, arc((0, 0), 4, 0, qlb))
```

— *"the arc of radius 4 through your angle: is it 10 cm long?"* That is how *find the angle* and *find the radius* get checked without any answer being stored.

One thing measuring cannot do: reject a root that the *situation* forbids. In task 10 both solutions of the quadratic make a genuine sector of the right area, and only the question knows that the angle was said to be acute. That mark is yours; the check will not withhold it. The same gap is flagged where it occurs.

A consequence worth having: **figures may be built from your own earlier answers.** In task 11 the width of the

gutter comes from the angle you found; in task 12 the height of the cone comes from the slant height you found. A wrong first part therefore costs you the first part only — which is what a markscheme calls follow through.

When you are wrong the check says **how**, and every diagnosis is built from the same contour:

what you wrote	what the check says
the polygon through the arc's ends	the arc has been read as a chord
the whole sector	a segment is the sector minus the triangle
the triangle	that is the triangle on the same chord
the arc alone	the boundary has straight pieces too
the other way round the circle	minor instead of major
one of several equal pieces	the figure has n of them
$180/\pi$ times the answer	the angle went in as degrees
the area of the cross-section	that is the section, not the solid
three times the answer	the factor of $\frac{1}{3}$ is missing

Order of work

level	what it means	tasks
	one measurement, one formula	1-3
	the shape has to be cut up first	4-8, 12-13
	the measurement is given and the shape is not	9-11

Every task is a real past-paper question, cited.

77% of these marks are on a calculator paper, and the number flatters it. Twelve of the thirteen tasks need nothing from the machine but arithmetic: there is no button for *the shaded part is a rectangle minus two segments*. The exception is task 6, where the equation $\theta - \sin \theta = c$ comes out and no algebra will solve it — and that equation, with the one like it in the timer, is 9 marks out of 105.

```

import sys
sys.path.append('.')          # from
practicum/geometry to practicum/kit.py
import sympy as sp           # the escape
hatch: anything not in kit is in sp
from kit import *             # checks +
sin, cos, pi, sqrt, arc, seg, cone, ...

language('en')                # this
notebook is in English, and so are the checks

th = symbols('theta')         # the letter
every one of these papers uses
DEGREE = pi / 180              # so that
210*DEGREE reads as '210 degrees'

# Every figure below may be drawn from an
answer you have not written yet.
# undrawn(...) returns an empty boundary in
that case, so the notebook runs
# top to bottom while it is still blank and
the checks print a white square.

def point(radius, angle, centre=(0, 0)):
    # the point of the circle at that angle
    if undrawn(radius, angle):
        return (Ellipsis, Ellipsis)
    return (centre[0] + radius * cos(angle),
            centre[1] + radius * sin(angle))

def sector(radius, start, end, centre=(0,
0)):
    # the boundary of a sector: out along one
radius, round, back along the other

```

```

        return undrawn(radius, start, end) or (
            seg(centre, point(radius, start,
centre)),
            arc(centre, radius, start, end),
            seg(point(radius, end, centre),
centre))

def segment(radius, start, end, centre=(0,
0)):
    # the boundary of a segment: round the
arc, back along the chord
    return undrawn(radius, start, end) or (
        arc(centre, radius, start, end),
        seg(point(radius, end, centre),
point(radius, start, centre)))

def fan(radius, angle, count):
    # count copies of the same sector, laid
round the centre one after another
    if undrawn(radius, angle):
        return undrawn(radius, angle)
    return [i for k in range(count) for i in
segment(radius, k * angle, (k + 1) * angle)]

print('ready; sympy', sp.__version__)
print('exact:      ', 3549 * pi / 16)
print('to 3 s.f.:  ', sig(3549 * pi / 16, 3))
print('a figure:   ', len(sector(4, 0,
Rational(5, 2))), 'pieces')

```

Map of the nine techniques

#	technique	you recognise it by	it reduces to
1	radians and arc length	two of s , r , θ are given	$s = r\theta$
2	angle against time	<i>one revolution every 16 seconds</i>	$\omega = 2\pi/T$, then technique 1
3	perimeter of a sector	the word <i>perimeter</i> beside an arc	walk the boundary and add
4	area of a sector	a shaded slice of pie	$\frac{1}{2}r^2\theta$
5	chord and half-angle	a chord, or a distance from the centre	drop a perpendicular; a right triangle
6	area of a segment	one straight side, not two	sector minus triangle
7	a composite region	the shape has no name	cut into 4, 5, 6 and add
8	two conditions, no numbers	neither r nor θ is given	eliminate, solve, reject a root

#	technique	you recognise it by	it reduces to
9	the circle leaves the plane	a cone	arc becomes circumference; Pythagoras

Techniques 1–4 are one idea applied four times. Technique 5 is C1 walking in: the perpendicular from the centre to a chord makes a right triangle, and everything about chords comes out of it. Technique 6 is the first subtraction and technique 7 is all the others. Technique 8 turns the topic round — you are given the measurement and asked for the shape — and technique 9 leaves the page.

Part 1. The arc *is* the angle

Theory: one definition, four consequences

Put an arc of length s on a circle of radius r . The angle it subtends at the centre, **measured in radians**, is defined as

$$\theta = \frac{s}{r} \quad \Longleftrightarrow \quad s = r\theta$$

Four things come straight out of it, and they are the whole of Part 1.

A full turn is 2π . The whole circumference is $2\pi r$, so $\theta = 2\pi r/r = 2\pi$. Hence $180^\circ = \pi$, and hence degrees convert by $\times \pi/180$.

A sector is a fraction of the disc. The slice at angle θ is $\theta/2\pi$ of the whole, so

$$A = \frac{\theta}{2\pi} \cdot \pi r^2 = \frac{1}{2} r^2 \theta$$

The perimeter of a sector is not the arc. Walk round the shaded slice with

your finger: out along a radius, round the arc, back along the other radius. Three pieces:

$$P = r\theta + 2r$$

The paper asks for this and marks it separately. May 2023 TZ2 gives it a whole mark of its own: "*arc + 2 × radius*".

The larger sector uses $2\pi - \theta$. If the shaded piece is the big one, the angle in the formula is the reflex angle, and reading it off the diagram is the only step that can go wrong.

The degrees trap is not a rounding error

Substituting degrees into $\frac{1}{2}r^2\theta$ does not give a slightly wrong answer; it gives one that is $180/\pi \approx 57$ times too big. The markscheme of November 2021 Paper 2 is explicit about the price:

"Award a maximum of M1A1A0A0A0 if a candidate uses degrees, even if later work is correct."

Two marks out of five for a completely correct method.

Task 1 ● — May 2023 TZ2 Paper 1
Q1, 6 marks

The following diagram shows a circle with centre O and radius 4 cm. The points P , Q and R lie on the circumference and $\widehat{POR} = \theta$, measured in radians. The shaded sector is POR , and **the length of arc PQR is 10 cm.**

(a) Find the perimeter of the shaded sector. **(b)** Find θ . **(c)** Find the area of the shaded sector.

Paper 1: no calculator, and none is needed. Notice that (a) can be answered before (b) — the arc is already given to you.

```
qla = ...           # the perimeter, in cm
qlb = ...           # the angle, in radians
qlc = ...           # the area, in cm^2

shaded = sector(4, 0, qlb)           #
your angle, drawn

verify_length('lb', 10, arc((0, 0), 4, 0,
qlb))    # is your arc 10 cm long?
verify_perimeter('la', qla, *shaded)
verify_area('lc', qlc, *shaded)
```

Task 2 — November 2025 TZ1 Paper 2 Q1, 6 marks

A regular pentagon is inscribed in a circle with centre O and radius r cm. The angle $A\hat{O}B$ is θ , and **arc AB is 12 cm**.

(a) Find **(i)** θ ; **(ii)** r . **(b)** Find the area of the shaded region — the five pieces of the disc left outside the pentagon.

The first check draws five arcs like yours, one after another. If your angle is right they close the circle; if it is not, they do not, and the check says so in those words.

```

q2th = ...      # the angle at the centre,
in radians
q2r  = ...      # the radius, in cm
q2b  = ...      # the shaded area, in cm^2

def rim(radius, angle):
    # five arcs like yours, laid one after
    another round the centre
    return undrawn(radius, angle) or [
        arc((0, 0), radius, k * angle, (k +
1) * angle) for k in range(5)]

verify_perimeter('2a', 5 * 12, *rim(q2r,
q2th))  # five arcs of 12 cm, right round
verify_area('2b', q2b, *fan(q2r, q2th, 5))

```

Task 3 — **May 2025 TZ1 Paper 2** **Q1, 6 marks**

The points A and B lie on a circle with centre O and radius 19.5 cm, such that $\widehat{BOA} = 210^\circ$. A piece of paper is cut into the shape of the sector BOA . A hollow cone with no base is then made from that sector by joining A to B ; the sector becomes the curved surface of the cone.

(a) Find the area of the sector BOA . **(b)** Find the radius of the cone.

Two different things happen to the same number here. In (a) the 210° becomes an angle in radians; in (b) the arc it cuts becomes a circle of a different radius. Nothing about the paper changes when you roll it: the cut edge keeps its length.

```
q3a = ...          # the area of the sector, in
cm^2
q3b = ...          # the radius of the cone's
base, in cm

CUT = 19.5 * 210 * DEGREE          # the curved
edge of the paper sector

verify_area('3a', q3a, *sector(19.5, 0, 210 *
DEGREE))
verify_length('3b', CUT, arc((0, 0), q3b, 0,
2 * pi))  # the same edge, joined up
```

● Part 2. Chords and segments

Theory: the perpendicular from the centre

Draw a chord $[AB]$ in a circle of radius r , with $\hat{AOB} = \theta$. Drop a perpendicular from O to the chord, meeting it at M . That single line does three things at once:

- it **bisects the chord**: $AM = MB$;
- it **bisects the angle**: $\hat{AOM} = \theta/2$;
- it makes a **right-angled triangle** OMA with hypotenuse r .

From that triangle, everything about chords:

$$\begin{aligned}AB &= 2r \sin \frac{\theta}{2}, \\OM &= r \cos \frac{\theta}{2}, \\ \theta &= 2 \arccos \frac{OM}{r}\end{aligned}$$

The cosine rule gives the first of these too — $AB^2 = r^2 + r^2 - 2r^2 \cos \theta$ — and the markscheme accepts either. The right triangle is usually shorter, and it is the only route when what you are given is the distance OM .

The mark that gets dropped here is the **doubling**. You find AM , or you find $\theta/2$, and then you write it down. May 2024 TZ2 gives "*recognizes that $AB = 2AM$* " its own mark.

Theory: a segment is a sector minus a triangle

The region between a chord and its arc has no formula of its own. It is a subtraction, and both parts you already have:

$$\underbrace{\frac{1}{2}r^2\theta}_{\text{sector } OAB} - \underbrace{\frac{1}{2}r^2 \sin \theta}_{\text{triangle } OAB} = \frac{1}{2}r^2(\theta - \sin \theta)$$

The triangle's area is $\frac{1}{2}ab \sin C$ from C1, with both sides equal to r .

Three numbers live in that one line — sector, triangle, difference — and the markscheme distinguishes them. So does the check: give it the sector and it will tell you that a segment is the sector minus the triangle; give it the triangle and it will say so.

The mixed-mode error. In $\frac{1}{2}r^2(\theta - \sin \theta)$ the letter θ appears twice: once as a number, once inside a sine. If your calculator is in degrees, the first is right and the second is wrong, and the answer is neither one thing nor the other. This is the quietest mistake in the topic.

Task 4 ● — *May 2022 TZ2 Paper 2*
Q1, 6 marks

The following diagram shows a circle with centre O and radius 5 metres. Points A and B lie on the circle and $\widehat{AOB} = 1.9$ radians. **The shaded sector is the larger one.**

(a) Find the length of the chord $[AB]$. **(b)** Find the area of the shaded sector.

```

q4a = ...          # the chord, in m
q4b = ...          # the shaded area, in m^2

A4, B4 = point(5, 0), point(5, 1.9)

verify_length('4a', q4a, seg(A4, B4))
verify_area('4b', q4b, *sector(5, 1.9, 2 *
pi))    # from B all the way round to A

```

Task 5 — May 2025 TZ3 Paper 2 Q4, 5 marks

Points A , B and C lie on a circle with centre O . The area of triangle AOB is 26 cm^2 and $\hat{AOB} = 2.51$ radians. C lies on the major arc.

Find the radius of the circle, and hence the length of arc ACB .

The paper asks only for the arc; the markscheme scores the radius on the way. Both are asked here, because a wrong radius and a wrong arc are two different mistakes and it is worth knowing which one you made.

```

q5r = ...          # the radius, in cm
q5b = ...          # the length of the major
arc ACB, in cm

def triangle(radius, angle):
    # the triangle AOB: two radii and the
    chord between their ends
    return undrawn(radius, angle) or (
        seg((0, 0), point(radius, 0)),
        seg(point(radius, 0), point(radius,
angle)),
        seg(point(radius, angle), (0, 0)))

verify_area('5 (radius)', 26, *triangle(q5r,
2.51))          # triangle AOB
verify_length('5 (arc)', q5b, arc((0, 0),
q5r, 2.51, 2 * pi))

```

Task 6 — May 2022 TZ1 Paper 2 Q2, 6 marks

A logo is created by removing two equal segments from a rectangle measuring 5 cm by 4 cm. The points A and B lie on a circle with centre O and radius 2 cm, such that $\widehat{AOB} = \theta$, where $0 < \theta < \pi$. One segment is removed from each of the two long sides.

(a) Find the area of one of the shaded segments **in terms of θ** . **(b)** Given that

the area of the logo is 13.4 cm^2 , find the value of θ .

Part (a) is checked by measuring the segment at three different angles, so any correct form passes. Part (b) is the one place in this notebook where the equation cannot be solved by hand — use a graph.

```

def one_segment(angle):
    # the segment cut off a circle of radius
    2 by a chord at that angle
    return segment(2, 0, angle)

def logo(angle):
    # the rectangle, 4 wide and 5 tall, with
    a segment bitten out of each side
    if undrawn(angle):
        return undrawn(angle)
    half = angle / 2
    away, high = 2 * cos(half), 2 * sin(half)
    # centre to chord, half-chord
    left, right = (-away, Rational(5, 2)), (4
    + away, Rational(5, 2))
    low_l, top_l = (0, Rational(5, 2) -
    high), (0, Rational(5, 2) + high)
    low_r, top_r = (4, Rational(5, 2) -
    high), (4, Rational(5, 2) + high)
    return (seg((0, 0), (4, 0)), seg((4, 0),
    low_r),
            arc(right, 2, pi + half, pi -
    half),
            seg(top_r, (4, 5)), seg((4, 5),
    (0, 5)), seg((0, 5), top_l),
            arc(left, 2, half, -half),
            seg(low_l, (0, 0)))

q6a = ...          # the area of one segment,
in terms of th
q6b = ...          # the angle, in radians

verify_law('6a', q6a, th, one_segment, (0.7,
1.9, 2.8))

```

```
verify_area('6b', 13.4, *logo(q6b))      #  
does your angle leave 13.4 cm^2?
```

Task 7 — November 2021 Paper 2 Q4, 6 marks

The following diagram shows a semicircle with centre O and radius r . Points P , Q and R lie on the circumference, such that $PQ = 2r$ and $\hat{ROQ} = \theta$, where $0 < \theta < \pi$. Two regions are shaded: the triangle POR , and the segment between the chord $[RQ]$ and its arc.

(a) Given that the areas of the two shaded regions are equal, show that $\theta = 2 \sin \theta$. **(b)** Hence determine the value of θ .

Part (a) is a show that — do it on paper, and the derivation is in the solutions. Part (b) starts from the equation (a) gives you, and the check scans the whole interval: a root is not enough, it has to be all of them.

```

q7 = [...]          # every value of theta in
(0, pi), as a list

# Three significant figures is what the paper
asks for, and a number
# rounded that far does not satisfy the
equation exactly. The tolerance
# is what the exam accepts, not what sympy
would.
verify_roots('7', q7, th - 2 * sin(th), (0.1,
pi), var=th, tol=1e-2)

```

Task 8 — May 2024 TZ1 Paper 1 Q3, 8 marks

Points A and B lie on the circumference of a circle of radius r cm with centre O . The angle $\widehat{AOB}$ is θ , measured in radians. **The perimeter of the sector is 10 cm and its area is 6.25 cm^2 .**

(a) Show that $4r^2 - 20r + 25 = 0$. **(b)** Hence, or otherwise, find the value of r and the value of θ .

Paper 1, and the last  before the shapes stop having names. Part (a) is a show that; part (b) is checked by drawing your sector and measuring it twice — its perimeter and its area are both given in the question.

```
q8r  = ...      # the radius, in cm
q8th = ...      # the angle, in radians

mine = sector(q8r, 0, q8th)

verify_perimeter('8 (perimeter)', 10, *mine)
verify_area('8 (area)', 6.25, *mine)
```

Part 3. The measurement is given; find the shape

Theory: cut the shape into pieces that have names

None of the shapes below is a sector, a segment or a triangle. All of them are sums and differences of those:

the shape	what it is made of
a logo: rectangle with two bites	rectangle – 2 segments
the letter C between two circles	sector of radius R – sector of radius r
a gutter's cross-section	segment + rectangle
a disc outside an inscribed pentagon	5 segments

Two rules make the cutting reliable.

Write each piece on its own line, with its own sign. The arithmetic is never the difficulty; keeping track of what has been added and what subtracted is.

Count the repeats before you finish.

Five segments, two bites, two vertical sides. The markscheme of November 2025 TZ1 gives a separate mark for the $\times 5$.

The markscheme of the gutter question is unusually frank about all this:

"There are many different ways to dissect the cross-section to determine its area."

They all give the same number, and choosing one is drawing, not calculating.

Theory: two conditions instead of two numbers

The hardest questions in the topic give you no r and no θ . They give you two *measurements* — a perimeter and an area, an area and a ratio — and both of those are expressions in r and θ . Two equations, two unknowns:

$$2r + r\theta = 10 \quad \text{and} \quad \frac{1}{2}r^2\theta = 6.25$$

The first one gives $r\theta = 10 - 2r$. Put that into the second, written as $\frac{1}{2}r \cdot (r\theta)$:

$$\frac{1}{2}r(10 - 2r) = 6.25 \implies 4r^2 - 20r + 25 = 0$$

Eliminate the one you are not asked for; solve what is left; and then do the step that carries its own mark:

Reject the root the question forbids. *"Award (A1)(M1)A0 if additional answers are given"* — November 2025 TZ3. The circle does not care that $\theta = 9.73$ solves the quadratic; the question said the angle was acute.

Sometimes what is left is not a quadratic at all. $20 - 4\theta + 4\sin\theta = 13.4$ has no algebraic solution, and none is expected: this is where the calculator finally does something a pencil cannot.

Task 9 — **November 2023 Paper 2 Q3, 7 marks**

A logo is a letter "C" formed between two circles with centre O . The point A lies on the inner circle, radius r cm,

where $r < 10$; the point B lies on the outer circle, radius 10 cm. **The reflex angle $\hat{AOB}$ is 5.2 radians**, and the C is the shaded region between the two arcs.

(a) Show that the area of the C is $260 - 2.6r^2$. **(b)** Given that the area of the C is 64 cm^2 , find **(i)** the value of r ; **(ii)** the perimeter of the C .

Part (a) is a show that. In (b) the figure is built from your own r , so if (i) is wrong then (ii) is judged against the C you actually drew.

```
def letter_c(inner):
    # the region between two arcs of 5.2
    radians, plus the two straight ends
    return undrawn(inner) or (
        seg((inner, 0), (10, 0)),
        arc((0, 0), 10, 0, 5.2),
        seg(point(10, 5.2), point(inner,
5.2)),
        arc((0, 0), inner, 5.2, 0))

q9r = ...          # the inner radius, in cm
q9p = ...          # the perimeter of the C, in
cm

verify_area('9 (radius)', 64, *letter_c(q9r))
verify_perimeter('9 (perimeter)', q9p,
*letter_c(q9r))
```

Task 10 — **November 2025 TZ3** **Paper 2 Q3, 6 marks**

Consider a circle of radius 8 mm and a sector of radius r mm whose acute angle at the centre is θ radians. **The perimeter of the sector is 1.5 times the circumference of the circle.**

(a) Show that $r = \frac{24\pi}{\theta + 2}$. **(b)** It is given that the area of the circle is the same as the area of the sector. Determine the value of θ .

The quadratic in (b) has two roots and only one of them is acute. Write down the acute one, and only that one.

Where the check is softer than the exam. Both roots build a real sector, and both sectors have exactly the area asked for — they are simply different sectors. Measuring cannot tell them apart, so the check will accept 9.73. The markscheme will not: *"Award (A1) (M1)A0 if additional answers are given."* That mark is yours to earn on your own.

```
def from_part_a(angle):
    # part (a) gives the radius, so your
    angle draws the whole sector
    return undrawn(angle) or sector(24 * pi /
    (angle + 2), 0, angle)

q10 = ...          # the angle, in radians

verify_area('10', pi * 8 ** 2,
*from_part_a(q10))  # equal to the circle
```

Task 11 ● — May 2023 TZ1 Paper 2
Q10(a)(b), 10 marks

A gutter is made by folding a sheet of metal 45 cm wide. Its cross-section is shaded in the diagram: arc AB lies on a circle with centre O and radius 12 cm, the two vertical sides above A and B are 10 cm high, and the width of the gutter is w cm. Let $\hat{AOB} = \theta$, where $0 < \theta < \pi$.

(a) Show that $\theta = 2.08$, correct to three significant figures. **(b)** Find the area of the cross-section of the gutter.

The whole 45 cm of metal is the boundary of the cross-section except the open top: up one side, round the arc, up the other. That is what the first check walks along. The second measures the shape your angle makes.

```

def corners(angle):
    # A and B at the ends of the arc, C and D
    ten centimetres above them
    low, high = -pi / 2 - angle / 2, -pi / 2
    + angle / 2
    A11, B11 = point(12, low), point(12,
high)
    return A11, B11, (B11[0], B11[1] + 10),
(A11[0], A11[1] + 10)

def gutter(angle):
    # the cross-section: the arc below AB,
    and the rectangle above it
    if undrawn(angle):
        return undrawn(angle)
    A11, B11, C11, D11 = corners(angle)
    return (arc((0, 0), 12, -pi / 2 - angle /
2, -pi / 2 + angle / 2),
        seg(B11, C11), seg(C11, D11),
seg(D11, A11))

def metal(angle):
    # the sheet: down one side, round the
    arc, up the other. The top is open.
    if undrawn(angle):
        return undrawn(angle)
    A11, B11, C11, D11 = corners(angle)
    return (seg(D11, A11), gutter(angle)[0],
seg(B11, C11))

q11a = ...          # the angle, in radians
q11b = ...          # the area of the cross-
section, in cm^2

```

```
verify_length('l1a', 45, *metal(q11a))  
verify_area('l1b', q11b, *gutter(q11a))
```

Part 4. The circle leaves the plane

Theory: a cone is a rolled-up sector

Cut a sector of radius l out of paper and join its two straight edges. The result is a cone with no base, and **nothing about the paper has changed**: the radius of the sector is now the slant height, and the arc is now the circle round the bottom.

$$\underbrace{l\theta}_{\text{the arc}} = \underbrace{2\pi R}_{\text{the base circle}}$$

That single equation is the whole of the conversion, and it is why the curved surface area of a cone is πRl and not something you have to remember: the sector's area is $\frac{1}{2}l^2\theta = \frac{1}{2}l \cdot l\theta = \frac{1}{2}l \cdot 2\pi R = \pi Rl$.

The rest is Pythagoras and one formula from the booklet:

$$l^2 = R^2 + h^2,$$

$$V = \frac{1}{3}\pi R^2 h,$$

$$S_{\text{total}} = \underbrace{\pi R^2}_{\text{base}} + \underbrace{\pi Rl}_{\text{curved}}$$

Two traps, and both are about what is included. *Total* surface area has the base in it; *curved* surface area does not. And the $\frac{1}{3}$ in the volume is the difference between a cone and the cylinder around it — the check will tell you if you have dropped it, in exactly those words.

The same solid on the two papers

The archive gives this technique on both papers, and the difference is only in the shape of the answer.

	May 2021 TZ2 Paper 1	May 2024 TZ1 Paper 2
given	$R = 2$, total surface 12π	$h = 20$, R from part (a)
work	$4\pi + 2\pi l = 12\pi$; $h =$ $\sqrt{16 - 4}$	substitute
answer	$\frac{8\sqrt{3}\pi}{3}$	851 m^3

Same formulas, same two lines of work. On Paper 1 a decimal earns nothing; on Paper 2 three significant figures are expected. The mathematics does not notice which paper it is on — but the markscheme does.

Task 12 — **May 2021 TZ2 Paper 1 Q10(c)(d) and May 2024 TZ1 Paper 2 Q2(b), 9 marks**

A right cone has a total surface area of 12π , base radius 2, height h and slant height l .

(a) Find the value of l . **(b)** Hence find the volume of the cone. *Give an exact answer.*

A monument is a right cone of vertical height 20 m. Part (a) of that question gives the base radius as 6.37 m.

(c) Find the volume of the monument.

The check in (a) unrolls the cone: a disc for the base, and the sector that the curved surface came from. In (b) and (c) the solid is spun out of its own cross-section, so no volume formula is stored anywhere.

```

q12a = ...      # the slant height l
q12b = ...      # the volume of the cone,
exactly
q12c = ...      # the volume of the
monument, in m^3

def unrolled(slant):
    # the cone opened out flat: the base
    disc, and the sector it came from
    return undrawn(slant) or (
        (arc((0, 0), 2, 0, 2 * pi),) +
        sector(slant, 0, 2 * pi * 2 / slant))

verify_area('l2a', 12 * pi, *unrolled(q12a))

verify_volume('l2b', q12b, *cone(radius=2,
slant=q12a), exact=True)
verify_volume('l2c', q12c,
*ccone(radius=6.37262, height=20))

```

Task 13 — **May 2024 TZ2 Paper 2 Q11(a)-(d), 9 marks**

A rotating sprinkler at a fixed point S waters every point inside a circle of radius 20 m. S is 14 m from the edge of a straight path, and that edge meets the circle at A and B .

(a) Show that $AB = 28.57$, correct to four significant figures. **(b)** The sprinkler makes one revolution every 16 seconds.

Show that it turns through $\frac{\pi}{8}$ radians in one second. **(c)** Let T seconds be the time for which $[AB]$ is watered in each revolution. Find the value of T . **(d)** At $t = 0$ the water crosses the path at A ; at time t it crosses at D . Write down an expression for $\alpha = \widehat{ASD}$ in terms of t .

(a), (b) and (d) print their own answers, so only (c) is checked — and it is checked with the two numbers (a) and (b) hand you. Your T seconds turn the jet through some angle; the chord it cuts must be the 28.57 m of part (a).

```
def jet(seconds):
    # the chord between where the jet crosses
    the path and where it leaves
    if undrawn(seconds):
        return undrawn(seconds)
    swept = seconds * pi / 8
    #
    the angle your T seconds turns through
    return (seg(point(20, -swept / 2),
    point(20, swept / 2)),)

q13c = ...          # the time, in seconds

verify_length('13c', 28.5657, *jet(q13c))
```

Trainer: name the technique in five seconds

Twelve openings. Do not compute anything — say only **which move you would make first**.

code	technique
arc	two of s , r , θ are known; use $s = r\theta$
rate	something turns; find $\omega = 2\pi/T$ first
perimeter	walk the boundary: arc plus the straight pieces
sector	$\frac{1}{2}r^2\theta$, with the right θ
chord	drop the perpendicular; a right triangle
segment	sector minus triangle
region	cut the shape into named pieces and add
unknown	two conditions, eliminate, solve, reject a root
cone	the arc becomes a circumference; Pythagoras

1. A sector of a circle of radius 4 cm has an arc of length 10 cm. Find the angle at the centre.
2. Find the perimeter of a sector of radius 4 cm whose arc is 10 cm long.

3. A circle has radius 5 m and $\angle AOB = 1.9$. Find the area of the larger of the two sectors.
4. A circle has radius 5 m and $\angle AOB = 1.9$. Find the length of the chord $[AB]$.
5. Find the area of the region between a chord $[AB]$ and its arc, given $r = 2$ and $\angle AOB = \theta$.
6. Find the area of a logo made by removing two equal segments from a 5 cm by 4 cm rectangle.
7. A sprinkler makes one revolution every 16 seconds. Find the angle it turns through in one second.
8. A paper sector of radius 19.5 cm is rolled up until its straight edges meet. Find the radius of the base.
9. The perimeter of a sector is 10 cm and its area is 6.25 cm^2 . Find r and θ .
0. Convert 210° into radians and find the area of the sector of radius 19.5 cm.
1. Find the volume of a right cone with base radius 2 and slant height 4.

2. The Earth is a sphere of radius 6000 km, and the shortest distance from M to N across its surface is 6000 km. Find the angle $\hat{MON}$.

```
answers = {
  1: '', 2: '', 3: '', 4: '', 5: '', 6: '',
  7: '', 8: '', 9: '', 10: '', 11: '', 12:
  '',
}

trigger_check(answers, {1: '404fcfb394d2', 2:
'6dc449e2bfbfbd', 3: '9d5d7fcf3aa3', 4:
'25735baaa5b4', 5: '03e71c6d7dc6', 6:
'c697d2981bf4', 7: 'c549779d79e5', 8:
'cff179e7721e', 9: 'b23a6a8439c0', 10:
'9d5d7fcf3aa3', 11: 'cff179e7721e', 12:
'404fcfb394d2'})
```

On the clock — *May 2025 TZ2* *Paper 2 Q6, 6 marks*

Six marks, nine minutes. No hints this time.

The following diagram shows a circle with centre O and radius r cm. Points A and B lie on the circle and $\hat{AOB} = \theta$ radians. The sector OAB is divided into

two regions: a shaded segment P and a triangle Q .

The area of the shaded segment P is 12.8 cm^2 , and the areas of P and Q are in the ratio $3 : 5$.

Find the value of θ and the value of r .

The paper asks only for r ; the markscheme finds θ first and scores it. Both are asked here.

Attempt log

date	time	result

```
qt_th = ...      # the angle, in radians
qt_r  = ...      # the radius, in cm

# the shaded segment is 12.8; segment and
# triangle together fill the
# sector, so the segment is three eighths of
it
verify_area('timer, segment', 12.8,
*segment(qt_r, 0, qt_th))
verify_area('timer, sector', 12.8 *
Rational(8, 3), *sector(qt_r, 0, qt_th))
```



Solutions

Work these only after you have your own answer, or you are reading, not practising.

1 (a) Walk the boundary: out along a radius, round the arc, back along the other radius. The arc is given as 10 cm:

$$P = 10 + 4 + 4 = \boxed{18 \text{ cm}}$$

The markscheme gives a mark for the phrase "*arc + 2 × radius*" alone. You never need θ for this part.

1 (b) $s = r\theta$ with $s = 10$ and $r = 4$:

$$10 = 4\theta \implies \theta = \boxed{\frac{5}{2}}$$

1 (c) $A = \frac{1}{2}r^2\theta = \frac{1}{2} \cdot 16 \cdot \frac{5}{2} = \boxed{20 \text{ cm}^2}$.

Or, without θ at all: $A = \frac{1}{2}rs = \frac{1}{2} \cdot 4 \cdot 10 = 20$. The two are the same identity read in two directions.

2 (a) A regular pentagon splits the full turn into five equal parts:

$$\theta = \frac{2\pi}{5} \approx 1.2566$$

and then $s = r\theta$ with $s = 12$:

$$12 = r \cdot \frac{2\pi}{5} \implies r = \frac{60}{2\pi} = \boxed{\frac{30}{\pi}} \approx 9.5493 \text{ cm}$$

2 (b) The shaded region is the five pieces of the disc left outside the pentagon, and each is a **segment** on a side of the pentagon:

$$\begin{aligned} 5 \times \frac{1}{2}r^2(\theta - \sin \theta) &= \frac{5}{2} \left(\frac{30}{\pi} \right)^2 \left(\frac{2\pi}{5} - \sin \frac{2\pi}{5} \right) \\ &= \boxed{69.7 \text{ cm}^2} \end{aligned}$$

The markscheme's other route subtracts the whole pentagon from the whole disc, $\pi r^2 - 5 \cdot \frac{1}{2}r^2 \sin \theta$, which is the same five segments collected differently. Either way, the mark that goes missing is the $\times 5$.

3 (a) 210° in radians is $210 \cdot \frac{\pi}{180} = \frac{7\pi}{6}$, so

$$A = \frac{1}{2}(19.5)^2 \cdot \frac{7\pi}{6} = \boxed{\frac{3549\pi}{16}} \approx 697 \text{ cm}^2$$

The degree form $\frac{210}{360}\pi r^2$ gives the same number and the same marks.

3 (b) Rolling the paper changes nothing about its edges. The cut edge was an arc of length

$$s = 19.5 \cdot \frac{7\pi}{6} = \frac{91\pi}{4} \approx 71.47 \text{ cm}$$

and after rolling it is the circle round the bottom of the cone:

$$2\pi R = \frac{91\pi}{4} \implies R = \boxed{\frac{91}{8}} = 11.375 \text{ cm}$$

The sector's own radius, 19.5, has become the **slant height**. Confusing it with R is the standard mistake here.

4 (a) Either the cosine rule,

$$\begin{aligned} AB^2 &= 5^2 + 5^2 - 2 \cdot 5 \cdot 5 \cos 1.9 \\ &\implies AB \\ &= \sqrt{50 - 50 \cos 1.9} \end{aligned}$$

or the perpendicular from O , which halves both chord and angle:

$$AB = 2 \cdot 5 \sin \frac{1.9}{2} = 10 \sin 0.95 = \boxed{8.13 \text{ m}}$$

Both are in the markscheme. The second is shorter and generalises.

4 (b) The shaded sector is the **larger** one, so its angle is the reflex angle:

$$\begin{aligned} A &= \frac{1}{2}(5)^2(2\pi - 1.9) \\ &= \frac{1}{2} \cdot 25 \cdot 4.3832 \\ &= \boxed{54.8 \text{ m}^2} \end{aligned}$$

Or take the whole disc and remove the small sector: $\pi(5)^2 - \frac{1}{2}(25)(1.9) = 78.54 - 23.75 = 54.79$. Reading 1.9 off the diagram and using it directly gives 23.8 — a complete method, no marks.

5 The triangle AOB has two sides equal to r and the angle 2.51 between them:

$$\begin{aligned}
\frac{1}{2}r^2 \sin 2.51 &= 26 \\
&\implies r^2 \\
&= \frac{52}{\sin 2.51} \\
&\implies r \\
&= \boxed{9.38 \text{ cm}}
\end{aligned}$$

C is on the major arc, so arc ACB uses the reflex angle:

$$\begin{aligned}
s &= r(2\pi - 2.51) \\
&= 9.3846 \times 3.7732 \\
&= \boxed{35.4 \text{ cm}}
\end{aligned}$$

Using 2.51 instead gives 23.6, which is the arc on the other side of the chord — the check names that mistake in those words.

6 (a) Sector minus triangle, with $r = 2$:

$$\frac{1}{2}(2)^2\theta - \frac{1}{2}(2)^2 \sin \theta = \boxed{2\theta - 2 \sin \theta}$$

6 (b) The rectangle is $5 \times 4 = 20$, and two of those segments are removed:

$$20 - 2(2\theta - 2 \sin \theta) = 13.4$$

$$20 - 4\theta + 4 \sin \theta = 13.4$$

There is no algebraic route from here — θ appears both inside and outside a sine — so graph $y = 20 - 4\theta + 4 \sin \theta$ against $y = 13.4$ on $0 < \theta < \pi$:

$$\theta = \boxed{2.36}$$

The markscheme's other opening is to note that one segment must be $(20 - 13.4)/2 = 3.3$ and solve $2\theta - 2 \sin \theta = 3.3$. Same equation.

"do not accept an answer in degrees", says the markscheme, and then names the number it will not accept: 135.030...

7 (a) $PQ = 2r$ makes $[PQ]$ a diameter, so P , O and Q are collinear and $\hat{POR} = \pi - \theta$.

The segment between chord $[RQ]$ and its arc:

$$\frac{1}{2}r^2\theta - \frac{1}{2}r^2 \sin \theta$$

The triangle POR has sides $OP = OR = r$ and the angle $\pi - \theta$ between them:

$$\frac{1}{2}r^2 \sin(\pi - \theta)$$

Set them equal, cancel $\frac{1}{2}r^2$, and use $\sin(\pi - \theta) = \sin \theta$:

$$\theta - \sin \theta = \sin \theta \implies \theta = 2 \sin \theta$$

7 (b) On $0 < \theta < \pi$ the equation $\theta = 2 \sin \theta$ has exactly one solution:

$$\theta = \boxed{1.90}$$

($\theta = 0$ solves it too, and is excluded by the question — a segment of zero angle is not a region.) The markscheme adds "*Award A0 if there is more than one solution*", which is why the check scans the whole interval rather than testing the one number you wrote.

8 (a) Two conditions, both in r and θ :

$$2r + r\theta = 10, \quad \frac{1}{2}r^2\theta = 6.25$$

From the first, $r\theta = 10 - 2r$. Substitute into the second, written as $\frac{1}{2}r(r\theta)$:

$$\begin{aligned}
\frac{1}{2}r(10 - 2r) &= 6.25 \\
&\implies 10r - 2r^2 \\
&= 12.5 \\
&\implies 4r^2 - 20r + 25 \\
&= 0
\end{aligned}$$

8 (b) That quadratic is a perfect square:

$$(2r - 5)^2 = 0 \implies r = \boxed{\frac{5}{2}}$$

A repeated root, so there is nothing to reject here. Back-substituting into the perimeter:

$$2 \cdot \frac{5}{2} + \frac{5}{2}\theta = 10 \implies \frac{5}{2}\theta = 5 \implies \theta = \boxed{2}$$

9 (a) Both arcs subtend the same reflex angle 5.2, so the C is one sector minus another:

$$\frac{1}{2}(10)^2(5.2) - \frac{1}{2}r^2(5.2) = 260 - 2.6r^2$$

9 (b)(i) $260 - 2.6r^2 = 64$, so

$$\begin{aligned}
 r^2 &= \frac{196}{2.6} \\
 &= \frac{980}{13} \\
 &\implies r \\
 &= \frac{14\sqrt{65}}{13} \\
 &= \boxed{8.68 \text{ cm}}
 \end{aligned}$$

The negative root is not a radius, and $r < 10$ was given.

9 (b)(ii) Walk the boundary of the C: the outer arc, the inner arc, and the **two straight ends** where the two circles are joined:

$$\underbrace{10(5.2)}_{52} + \underbrace{r(5.2)}_{45.15} + \underbrace{2(10 - r)}_{2.635} = \boxed{99.8 \text{ cm}}$$

Leaving out the straight ends gives 97.1, and the check says so by name.

10 (a) The circle has circumference $2\pi(8) = 16\pi$, so the perimeter of the sector is $1.5 \times 16\pi = 24\pi$. A sector's perimeter is $r\theta + 2r$:

$$\begin{aligned}
r\theta + 2r &= 24\pi \\
&\implies r(\theta + 2) \\
&= 24\pi \\
&\implies r \\
&= \frac{24\pi}{\theta + 2}
\end{aligned}$$

10 (b) Equal areas:

$$\pi(8)^2 = \frac{1}{2}r^2\theta = \frac{1}{2}\left(\frac{24\pi}{\theta + 2}\right)^2\theta$$

$$64\pi(\theta + 2)^2 = 288\pi^2\theta \implies (\theta + 2)^2 = 4.5\pi\theta$$

$$\theta^2 + (4 - 4.5\pi)\theta + 4 = 0$$

$$\implies \theta$$

$$= 0.411 \text{ or } 9.73$$

The angle was said to be **acute**, so

$$\theta = \boxed{0.411}$$

Writing both down costs the last mark:
"Award (A1)(M1)A0 if additional answers are given."

11 (a) The metal is 45 cm wide and becomes two vertical sides and the arc:

$$45 = 10 + s + 10 \implies s = 25$$

and $s = r\theta$ with $r = 12$:

$$\theta = \frac{25}{12} = 2.0833\dots = \boxed{2.08} \text{ (3 s.f.)}$$

11 (b) Cut the cross-section into the segment below $[AB]$ and the rectangle above it. First the width, from the half-angle:

$$w = 2 \cdot 12 \sin \frac{2.08}{2} = 20.70 \text{ cm}$$

Then the segment:

$$\begin{aligned} \frac{1}{2}(12)^2(2.08 - \sin 2.08) &= 149.76 - 62.87 \\ &= 86.89 \text{ cm}^2 \end{aligned}$$

and the rectangle $10w = 206.98 \text{ cm}^2$:

$$86.89 + 206.98 = 293.9 \approx \boxed{294 \text{ cm}^2}$$

Keeping $\theta = 25/12$ throughout gives 294.4 — the same to three significant figures, and the markscheme accepts both. The markscheme also says outright that *"there are many different ways to dissect the cross-section"*: two trapezia plus a

sector is one of them, and it lands on the same number.

12 (a) Total surface area is the base plus the curved part:

$$\begin{aligned}\pi r^2 + \pi r l &= 12\pi \\ \implies 4\pi + 2\pi l & \\ &= 12\pi \\ \implies 2\pi l & \\ &= 8\pi \\ \implies l & \\ &= \boxed{4}\end{aligned}$$

Forgetting the base gives $\pi \cdot 2 \cdot l = 12\pi$ and $l = 6$.

12 (b) Pythagoras first, then the volume:

$$\begin{aligned}h &= \sqrt{l^2 - r^2} = \sqrt{16 - 4} = 2\sqrt{3} \\ V &= \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(4)(2\sqrt{3}) = \boxed{\frac{8\sqrt{3}\pi}{3}}\end{aligned}$$

Paper 1: 14.5 earns nothing.

12 (c) Nothing new — the same formula with h given:

$$V = \frac{1}{3}\pi(6.3726)^2(20) = 850.54 \approx \boxed{851 \text{ m}^3}$$

Paper 2: $\frac{8\sqrt{3}\pi}{3}$ would be fine too, but three significant figures are what is asked for. The markscheme also accepts 850, from the rounded $r = 6.37$.

13 (a) Let M be the midpoint of $[AB]$. Then $SM \perp AB$, $SM = 14$ and $SA = 20$:

$$\begin{aligned} AM &= \sqrt{20^2 - 14^2} \\ &= \sqrt{204} \\ &\implies AB \\ &= 2\sqrt{204} \\ &= 28.5657\dots \\ &= \boxed{28.57} \end{aligned}$$

13 (b) One revolution is 2π and takes 16 seconds:

$$\omega = \frac{2\pi}{16} = \boxed{\frac{\pi}{8}} \text{ rad s}^{-1}$$

13 (c) The half-angle at the centre comes from the same right triangle:

$$\begin{aligned}
\cos \frac{A\hat{S}B}{2} &= \frac{14}{20} \\
&\implies A\hat{S}B \\
&= 2 \arccos(0.7) \\
&= 1.5908
\end{aligned}$$

and time is angle divided by rate:

$$T = \frac{1.5908}{\pi/8} = \frac{16 \arccos 0.7}{\pi} = \boxed{4.05 \text{ s}}$$

13 (d) The angle grows at a constant $\pi/8$ per second:

$$\alpha = \boxed{\frac{\pi}{8}t}$$

Timer

P is the segment and Q the triangle, and together they are the sector. From $P : Q = 3 : 5$ with $P = 12.8$:

$$\begin{aligned}
Q &= \frac{5}{3}(12.8) \\
&= \frac{64}{3}, & P + Q \\
&= \frac{102.4}{3} \\
&= 34.133
\end{aligned}$$

Now write both areas in r and θ :

$$\frac{1}{2}r^2\theta = 34.133, \quad \frac{1}{2}r^2 \sin \theta = \frac{64}{3}$$

Divide one by the other and r disappears:

$$\frac{\theta}{\sin \theta} = \frac{102.4/3}{64/3} = 1.6 \implies \theta = 1.6 \sin \theta$$

which is the same shape of equation as task 7, and is solved the same way:

$$\theta = \boxed{1.599}$$

Then back into the segment:

$$\frac{1}{2}r^2(\theta - \sin \theta) = 12.8 \implies r = \boxed{6.53 \text{ cm}}$$

The ratio is the whole difficulty. Everything after it is two formulas you have used a dozen times, and the division that kills r is the move worth remembering: **when two areas of the same circle are given in a ratio, the radius cancels.**